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| PROJECT  | Credit Card Approval Prediction | TEAM | AI/ML Program |
| DOCUMENT | Solution Requirements           | DATE | July 2026     |

## Solution Requirements

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### Functional Requirements

- The system shall accept applicant demographic and financial details through a web form
- The system shall preprocess and encode inputs to match the trained model's expected format
- The system shall return a binary prediction: Approved or Rejected
- The system shall display the result on a dedicated results page
- The system shall be trained on historical applicant and credit history data

### Non-Functional Requirements

- The prediction shall be returned to the user within a few seconds (real-time)
- The web interface shall be simple and usable without technical knowledge
- The model training pipeline shall be reproducible via a documented notebook
- The application shall run on a standard local development server

### Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB, recommended 8 GB
- Storage: Minimum 2 GB free disk space
- Internet connection: required for dataset download, package installation, and cloud deployment

### Software Requirements

- Operating System: Windows / Linux / macOS
- Python 3.8 or above (Python 3.10 used for local deployment)
- Google Colab / Jupyter Notebook for data analysis and model training
- VS Code for Flask application development
- Web browser (Chrome / Edge / Firefox) for testing